Northern Devon Healthcare NHS Trust

Document Control

Title
Arterial Line Removal for Neonates Standard Operating Procedure (including peripheral and umbilical lines)

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Document Class Standard Operating Procedure	**Target Audience** Neonatal Staff
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Superseded Documents
Arterial Line Removal for Neonates Standard Operating Procedure v1.0

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January 2018	January 2021	Three years

Consulted with the following stakeholders: • Neonatal Lead Consultant • Senior Neonatal Nurses	**Contact responsible for implementation and monitoring compliance:** Neonatal Lead Consultant **Education/ training will be provided by:** Neonatal Lead Consultant

Approval and Review Process
- Paediatric Specialty Team

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1. Purpose

This Standard Operating Procedure (SOP) has been written to:

- Identify the procedure for the safe removal of neonatal peripheral arterial lines and umbilical arterial lines
- reduce the risks associated with arterial line removal

2. Scope

This SOP relates to the following staff groups who may be involved in arterial line removal:

- Medical staff
  Registered neonatal nurses

3. Responsibilities

3.1. Role of Paediatrics Medical Team:

The Paediatric Medical Team are responsible for:

- Acting in accordance with this SOP
- Ensuring that risks associated with arterial line removal are minimised.

3.2. Role of Neonatal Staff:

The Neonatal Staff nurses are responsible for:

- Assisting the doctor in the procedure
- Observing the baby for associated risks and taking appropriate action

4. Indication

- When infants condition is such that the line is not required
- When complications occur that indicate the line should be removed

5. Definitions

UAC - Umbilical Arterial Catheter

UVC - Umbilical Venous Catheter

6. Procedure

Notes

- Removal of lines should occur as soon as no longer required
- The procedure is performed by medical staff.
- Do not remove arterial lines if alone in clinical area
- The procedure may require two people to support the infant during removal of line. Ideally the procedure should be performed to coincide with patient care although there may be occasions when urgent removal is required.
- Keep parents informed throughout procedure

6.1. Procedure for withdrawal of Peripheral Arterial Catheter

Steps	Procedure for Withdrawal of Peripheral Arterial Catheter
1	Prepare Equipment and follow aseptic non-touch technique • Sterile dressing pack (gauze, gloves, gallipot) • Specimen pot and sterile scissors • Tape
2	Ensure adequate analgesia
3	Remove dressing (care not to remove line)
4	Withdraw peripheral arterial catheter and place tip in specimen pot
5	Apply pressure to site **for 5 minutes** with sterile gauze ensuring circulation to hand/foot is maintained.
6	If oozing/bleeding continues continue applying pressure for further 5 minutes **before** checking again and inform medical staff
7	Cover site with a small piece of sterile gauze and tape
8	Maintain visibility of site for observation of further bleeding, bruising and perfusion of distal extremities for 4 hours post removal.

9	Sent catheter tip for microscopy
10	Recalculate fluid requirements and adjust other infusion rates following cessation of arterial infusion
11	Document that peripheral arterial line has been removed and any complications in the following places: 1. Medical notes 2. Nursing care plan

6.2. Procedure for Withdrawal of Umbilical Catheter

Steps	Procedure for Withdrawal of Umbilical Arterial and Venous Catheter
1	Prepare Equipment and follow aseptic non-touch technique • Sterile dressing pack, (gauze, gloves, gallipot) • Unisept sachets - Chlorhexidine gluconate 0.05% (use saline in extreme preterms) • Sterile stitch cutter • Sterile nylon tape • Sterile scissors and specimen pot • Tape
2	For UAC removal turn heparin infusion pump off and clamp infusion line. 30 mins prior to removal
3	Clearly identify line for removal prior to cutting sutures
4	Clean umbilical stump with unisept
5	Place nylon tape around umbilical stump
6	Cut stitches and tape attaching catheters
7	Slowly pull back the catheter about 0.5cm at a time ensuring that the artery is not pulsating to the last 5cm
	If the artery is pulsating wait a few minutes and start withdrawing the catheter again
8	Very slowly remove last 5cm of catheter
9	Apply a pressure dressing of folded sterile gauze to the umbilicus.
10	If no bleeding occurs after several minutes apply a dry gauze dressing.

11	If bleeding occurs continue to apply pressure
12	If it bleeds very heavily pull the nylon tape tightly around the base of the umbilical stump
13	Dispose of waste according to Trust policy
14	Sent catheter tip for microscopy
	Do not lie baby prone for at least 4 hours after removal of UAC/UVC. Check frequently for ooze/bleeding.
15	Ensure a nappy is secured below the umbilicus.
16	Recalculate fluid requirements and adjust other infusion rates with removal of lines
17	Document that peripheral arterial line has been removed and any complications in the following places: 1. Medical notes 2. nursing care plan

General notes for removal of umbilical catheters

- Only remove lines if umbilical stump is clearly visible
- If the umbilical cord has become very dry and appears to stick to the catheter place sterile gauze soaked in 0.9%saline around the umbilical stump to moisten it for easier removal of catheter. Leave cord to soak for about 20 minutes.
- A UAC should not remain in situ for more than 10 days unless specified by the Paediatric Consultant

7. Cross References

- ABL 90 Blood gas analyser competency
- Arterial blood sampling for neonates SOP
- Umbilical catheter insertion and management guidelines.
- Waste policy

8. References

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